FastVGGT: Training-Free Acceleration of Visual Geometry Transformer

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Abstract

Visual Geometry Grounded Transformer (VGGT) achieves state-of-the-art feed-forward 3D reconstruction from multi-view images, yet its quadratic-time global attention becomes a severe bottleneck when processing long image sequences, limiting both speed and memory scalability. We observe that global attention maps in VGGT exhibit strong similarity across tokens—a phenomenon we term token collapse*—indicating substantial computational redundancy. Based on this observation, we propose* ***FastVGGT****, a training-free token merging method that accelerates VGGT by merging redundant tokens during global attention. Our approach partitions tokens into three groups: first-frame reference tokens that serve as world anchors, salient tokens selected via fixed-stride sampling, and remaining tokens assigned as merge sources or destinations through region-based random sampling. Redundant tokens are merged via cosine-similarity-based averaging and unmerged before dense prediction heads. Combined with a memory-efficient variant (VGGT*) that discards unused intermediates, FastVGGT enables single-pass inference on sequences exceeding 1,000 frames. Experiments demonstrate up to* $4\times$ *speedup with equal or improved accuracy across point cloud reconstruction and camera pose estimation benchmarks, while mitigating long-sequence drift in pose estimation.*

1. Introduction

Reconstructing 3D scene geometry and estimating camera poses from image sequences is a fundamental problem in computer vision, with applications spanning autonomous driving, augmented reality, and robotics. Classical approaches based on Structure-from-Motion (SfM) [14] and visual SLAM [6, 10] achieve high accuracy but rely on iterative optimization pipelines that are computationally expensive and sensitive to initialization. Recent feed-forward methods [8, 20, 21, 23] have emerged as efficient alternatives, directly regressing 3D outputs from image pairs or sets using transformer architectures.

Among these, Visual Geometry Grounded Transformer (VGGT) [19] stands out as a unified model that jointly predicts camera parameters, point maps, depth maps, and 3D point tracks from arbitrary numbers of input views in a single forward pass. However, VGGT employs global attention across all image tokens at multiple transformer layers, resulting in quadratic computational complexity with respect to the total number of tokens. As shown in Fig. 2, global attention dominates inference time, accounting for the majority of computation. When scaling to long image sequences—a scenario of critical practical importance for video-based reconstruction and SLAM—this quadratic cost leads to prohibitive inference times and VRAM consumption, limiting VGGT to relatively short sequences in practice.

To understand this bottleneck, we analyze the global attention maps within VGGT and make a key observation: attention weights across tokens exhibit remarkably high similarity, a phenomenon we term *token collapse* (Fig. 3). This indicates that many tokens carry near-identical information after global attention, suggesting that a significant portion of the computation is redundant. This observation motivates a natural question: can we exploit this redundancy to accelerate VGGT without sacrificing reconstruction quality?

We answer this question affirmatively with **FastVGGT**, a training-free token merging strategy tailored to VGGT's global attention mechanism. Unlike prior token reduction methods [2, 3, 13] that typically require task-specific fine-tuning, FastVGGT operates as a plug-in module that merges redundant tokens before global attention and unmerges them afterward, requiring no retraining or architectural modification. Our method introduces a principled token partitioning strategy with three groups: (1) first-frame tokens are preserved as reference anchors that establish the world coordinate frame and are never merged; (2) salient tokens are selected from each subsequent frame via fixed-stride sampling to maintain spatial coverage; and (3) remaining tokens are assigned as merge sources or destinations through region-based random sampling. Token merging is performed via cosine-similarity-based averaging, and merged tokens are unmerged before dense prediction heads to restore full spatial resolution.

We further introduce **VGGT***, a memory-efficient variant that discards intermediate outputs not required for the target task, substantially reducing peak VRAM usage. Combined with our token merging strategy, FastVGGT

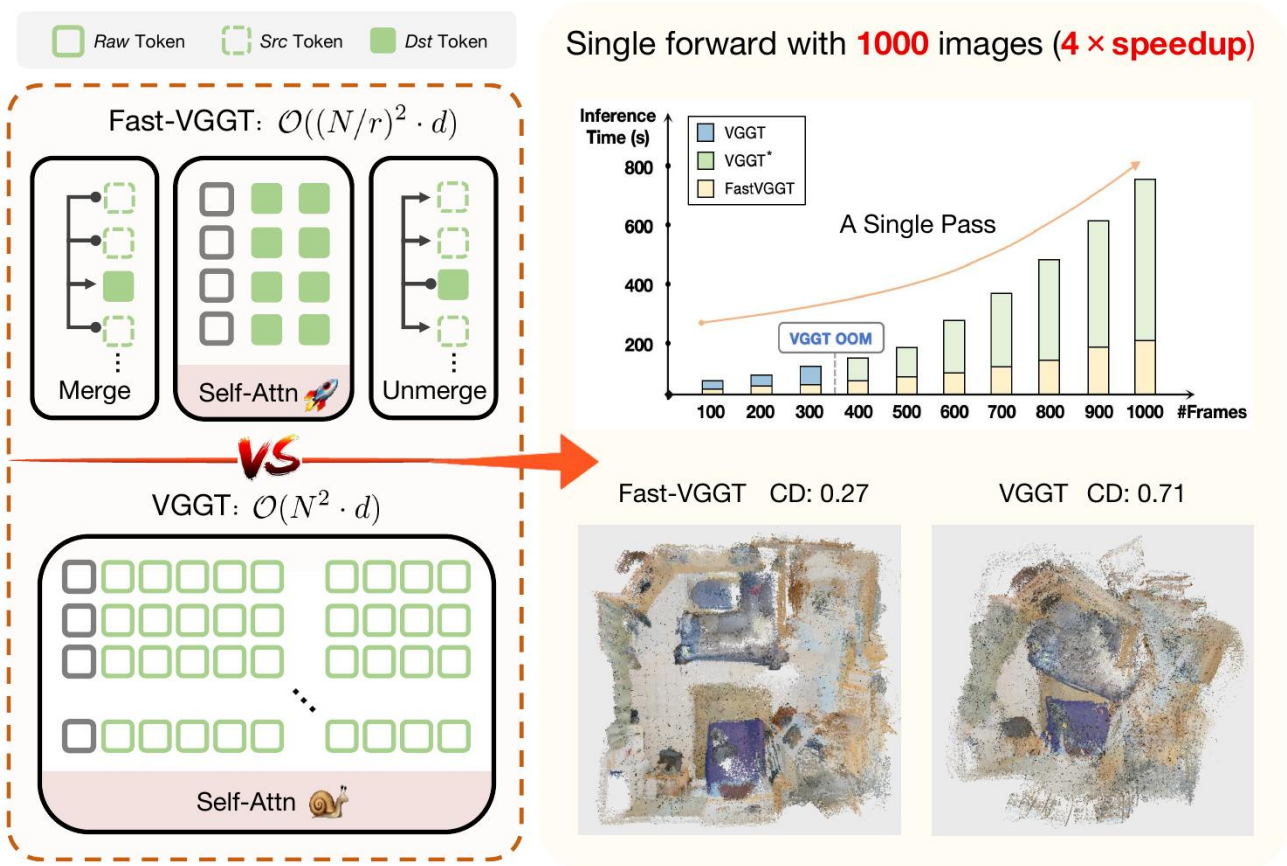

Figure 1. We propose **FastVGGT**, a training-free token merging framework for VGGT that processes 1,000 images in a single inference pass, achieving up to $4\times$ faster inference while mitigating long-sequence error accumulation. By merging redundant tokens in global attention with a 3D-aware partitioning strategy, FastVGGT scales to previously infeasible sequence lengths without sacrificing reconstruction or pose accuracy.

enables VGGT to process sequences of over 1,000 images in a single forward pass (Fig. 1), a regime previously infeasible. Extensive experiments demonstrate that FastVGGT achieves up to $4\times$ speedup while maintaining or even improving accuracy. On ScanNet with 1,000 images, FastVGGT reduces Chamfer Distance from 0.471 to 0.425 and inference time from 724.6s to 180.7s. Notably, token merging acts as an implicit regularizer that mitigates error accumulation in long-sequence pose estimation, reducing Absolute Trajectory Error from 0.196 to 0.164 (Fig. 5).

Our contributions are summarized as follows:

- We identify the *token collapse* phenomenon in VGGT's global attention, revealing substantial computational redundancy in long-sequence 3D reconstruction.
- We propose FastVGGT, a training-free token merging method with a principled partitioning strategy that preserves reference-frame anchors and spatial coverage while merging redundant tokens.
- We demonstrate that FastVGGT achieves up to $4\times$ speedup with equal or improved accuracy across multiple benchmarks, and enables single-pass inference on 1,000+ frame sequences.
- We show that token merging mitigates long-sequence drift, improving pose estimation accuracy on extended sequences.

2. Related Work

2.1. Feed-Forward 3D Reconstruction

Classical 3D reconstruction pipelines based on SfM [14] rely on iterative feature matching, triangulation, and bundle adjustment, which are accurate but computationally expensive and difficult to parallelize. DUSt3R [21] introduced a paradigm shift by casting pairwise reconstruction as direct pointmap regression using a transformer architecture, eliminating the need for explicit camera calibration. MASt3R [8] extended this approach with grounded feature

matching for improved multi-view consistency. However, these pairwise methods require costly global alignment when processing more than two views. MV-DUSt3R [17] addressed this by processing multiple views simultaneously in a single forward pass through multi-view decoder blocks. VGGT [19] further unified multiple 3D vision tasks—camera estimation, depth prediction, point cloud reconstruction, and tracking—into a single feed-forward transformer with global attention across all views. Fast3R [23] proposed an alternative multi-view architecture that processes over 1,000 images in one pass but focuses on reconstruction without the unified multi-task capability of VGGT. CUT3R [20] introduced continuous 3D perception for streaming inputs, while π^3 [22] explored probabilistic formulations for reconstruction under uncertainty. StreamVGGT [25] extends VGGT for streaming settings but does not address the computational bottleneck of global attention itself. FLARE [24] and Deng *et al.* [5] have explored related directions for accelerating or extending feed-forward reconstruction. Our work is complementary to these architectural advances: FastVGGT accelerates VGGT's global attention through token merging without modifying the underlying architecture.

2.2. Token Reduction and Merging

Token reduction methods aim to decrease the computational cost of vision transformers by removing or consolidating redundant tokens. TokenLearner [13] learns to select a compact set of informative tokens through spatial attention, but requires end-to-end training. Token Merging (ToMe) [1, 2] introduced a training-free approach that progressively merges similar tokens within ViT layers using bipartite matching, demonstrating that significant redundancy exists in image token representations. PUMER [3] proposed progressive token merging with task-specific importance scores. In the video domain, ViDTo [4] applied token downsampling to video transformers, while TempMe [15] exploited temporal redundancy for token merging across frames. Lee *et al.* [7] explored video token pruning, and Tran *et al.* [18] investigated acceleration strategies for video understanding models. SG-Former [12] proposed grouped attention as an alternative to full attention for efficiency. Shen *et al.* [16] introduced evolving token merging strategies. While these methods were designed for classification or video understanding tasks, applying token merging to 3D vision transformers with dense prediction outputs poses unique challenges: merged tokens must be faithfully unmerged to preserve spatial resolution for pointmap and depth estimation. FastVGGT adapts the token merging paradigm to this setting with a partitioning strategy that accounts for the geometric structure of multi-view reconstruction.

2.3. Visual Localization and Pose Estimation

Visual localization and camera pose estimation are core components of 3D scene understanding. Classical approaches include feature-based visual SLAM systems such as ORB-SLAM [9, 10], which track sparse keypoints across frames and jointly optimize camera poses and a sparse 3D map through local and global bundle adjustment. Direct methods such as LSD-SLAM [6] operate on pixel intensities rather than features, enabling semi-dense reconstruction. SfM systems like COLMAP [14] achieve high-quality reconstruction through incremental registration and optimization but scale poorly to long sequences. These iterative methods, while accurate, suffer from error accumulation over long trajectories and require loop closure or global optimization to maintain consistency. Feed-forward approaches such as VGGT [19] offer an alternative by directly regressing camera poses, but their accuracy can degrade on long sequences due to the lack of explicit global consistency constraints. Our experiments reveal that FastVGGT's token merging, by acting as an implicit regularizer on the attention computation, mitigates this drift and improves pose estimation accuracy on long sequences compared to the unmodified VGGT.

3. Method

3.1. Preliminary: VGGT Architecture

VGGT [19] is a feed-forward transformer that takes N unposed images as input and jointly predicts camera parameters, depth maps, and dense 3D point clouds. The architecture consists of four stages. First, a DINOv2 [11] backbone extracts patch-level features from each input image, producing a set of L tokens per frame. Second, *frame-level self-attention* layers process each frame independently, refining local features within each image. Third, *global alternating attention* layers operate across all N frames simultaneously, enabling cross-frame correspondence reasoning. Finally, task-specific dense prediction heads decode per-pixel outputs such as depth and point maps from the refined token representations.

The global attention stage is the primary computational bottleneck: it operates over the full set of $N \times L$ tokens, incurring $O(N^2L^2)$ complexity in both time and memory. As shown in Fig. 2, global attention accounts for the dominant share of total inference time, and its cost grows quadratically with the number of input frames, severely limiting scalability to long image sequences.

3.2. Motivation: Attention Redundancy

To understand whether the full token set is necessary during global attention, we visualize the attention maps at each stage of VGGT's global attention blocks. Specifically, we extract the attention weights for six representative tokens—

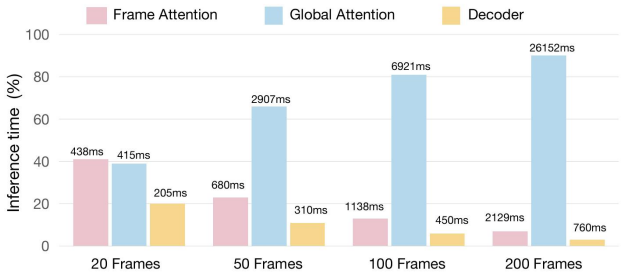

Figure 2. Component-wise analysis of VGGT inference time. Global attention dominates computation and grows quadratically with the number of input frames, motivating our token merging approach.

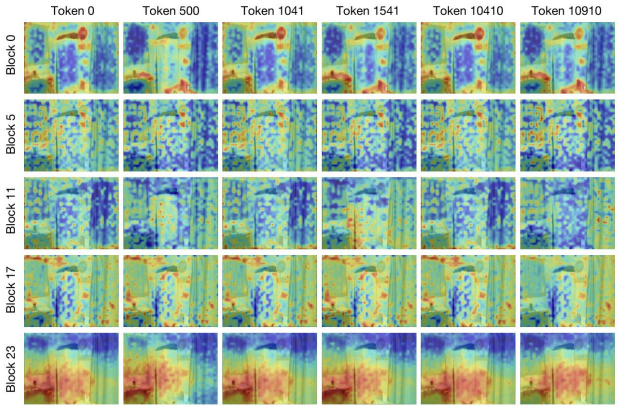

Figure 3. Visualizations of global attention maps in VGGT using six representative tokens (including the camera token and several image tokens). At every stage, different tokens exhibit strikingly similar attention patterns, revealing massive redundancy that FastVGGT exploits via training-free token merging.

including the camera token and several spatially distributed image tokens—and compare their attention distributions across all other tokens.

As shown in Fig. 3, the attention patterns of different tokens exhibit a striking degree of similarity at every stage of the network. Regardless of a token's spatial location or semantic role, it tends to attend to nearly the same set of tokens with similar weights. This phenomenon, which we term *token collapse*, reveals that global attention in VGGT carries massive redundancy: many tokens contribute nearly identical information to the attention computation.

This observation directly motivates a token merging strategy. If distinct tokens produce highly correlated attention distributions, then a carefully chosen subset of representative tokens can approximate the full attention output with minimal loss of information. Crucially, such merging can be applied at inference time without any retraining, since it exploits inherent redundancy in the pretrained model's representations rather than modifying the learned weights.

3.3. FastVGGT: Training-Free Token Merging

Building on the redundancy observation, we propose FastVGGT, a training-free acceleration framework that introduces 3D-aware token merging into VGGT's global attention blocks. An overview of the approach is illustrated in Fig. 4.

3D-Aware Token Partitioning. A naive token merging strategy that treats all tokens uniformly would disrupt the geometric consistency required for accurate 3D reconstruction. We therefore design a 3D-aware partitioning scheme that classifies tokens into three categories:

1. **Reference tokens.** All L tokens from the first frame are designated as *destination* (dst) tokens and are never merged. The first frame serves as the world coordinate anchor in VGGT's coordinate system; preserving all of its tokens is essential to maintain the global reference frame. Our ablation study confirms that removing this constraint causes catastrophic degradation in pose estimation (ATE increases from 0.149 to 0.722).
2. **Salient tokens.** For each non-first frame, approximately 10% of tokens are selected via fixed-stride spatial sampling. These tokens are protected from merging (`enable_protection`=True) to preserve fine-grained spatial correspondences across frames, which are critical for accurate depth and point cloud prediction.
3. **Region-based src/dst tokens.** The remaining tokens in each non-first frame are partitioned into source (src) and destination (dst) sets using a region-based bipartite strategy. We divide the spatial grid into non-overlapping regions of size $s_x \times s_y$, and randomly sample one dst token from each region to ensure uniform spatial coverage. All other tokens in these regions become src tokens, which are candidates for merging.

Token Merging. Given the src/dst partition, we merge source tokens into their most similar destination tokens. Specifically, we compute the cosine similarity between each src token and all dst tokens:

$$\mathrm{sim}(\mathbf{x}_i^{\mathrm{src}}, \mathbf{x}_j^{\mathrm{dst}}) = \frac{\mathbf{x}_i^{\mathrm{src}} \cdot \mathbf{x}_j^{\mathrm{dst}}}{\|\mathbf{x}_i^{\mathrm{src}}\| \, \|\mathbf{x}_j^{\mathrm{dst}}\|}, \tag{1}$$

where $\mathbf{x}_i^{\mathrm{src}}$ and $\mathbf{x}_j^{\mathrm{dst}}$ are the feature vectors of the i-th source and j-th destination token, respectively. Each source token is assigned to the destination token with the highest cosine similarity. The merge operation aggregates assigned source tokens into their destination via mean pooling, implemented efficiently using `scatter_reduce` with "mean" mode. To handle the large token sets, the similarity computation

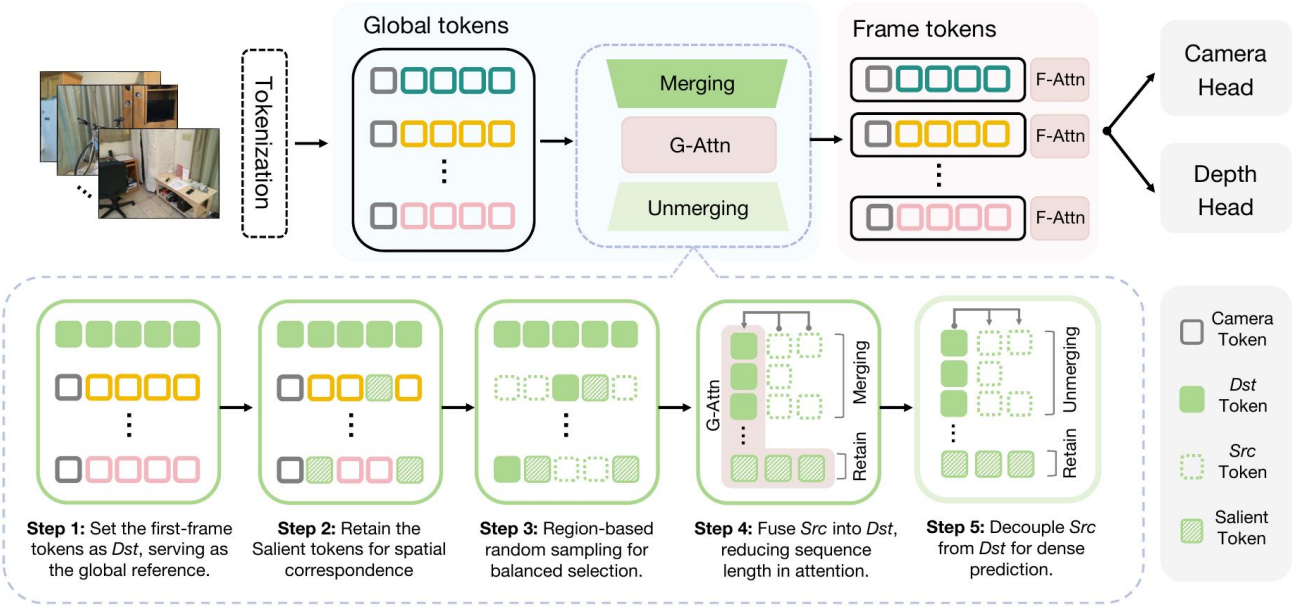

Figure 4. Overview of FastVGGT's training-free token merging strategy. Tokens are partitioned into three groups: reference tokens from the first frame (world anchors, never merged), salient tokens per subsequent frame (protected via fixed-stride sampling), and region-based src/dst tokens (merged via cosine-similarity matching). After global attention, an unmerging step restores full spatial resolution before dense prediction.

is performed via chunked matrix multiplication for memory efficiency.

This merging is applied within the global attention blocks of VGGT. With a merge ratio r (the fraction of src tokens to merge), a ratio of $r = 0.9$ across all global attention blocks provides the best trade-off between speed and accuracy.

Token Unmerging. Before the dense prediction heads, merged tokens must be restored to the original spatial resolution to produce per-pixel outputs. The unmerge operation scatters the merged destination token values back to all original token positions that were assigned to each destination during merging. This ensures that every spatial location receives a prediction, maintaining compatibility with the downstream heads for depth, point maps, and camera parameters.

VGGT*: Memory-Efficient Variant. To further reduce VRAM consumption for very long sequences, we introduce VGGT*, a memory-efficient variant that discards intermediate activations that are not required for subsequent computation. Combined with token merging, VGGT* enables inference on sequences of 1,000+ frames on a single GPU without running out of memory. In contrast, existing competitors run out of memory even on 500-frame inputs under identical hardware constraints.

4. Experiments

4.1. Experimental Setup

We evaluate FastVGGT on three benchmark datasets for indoor 3D reconstruction and camera pose estimation. **ScanNet-50** consists of 50 indoor scenes, from which we sample input sequences of 1000, 500, 300, and 100 images to assess scalability. **7 Scenes** provides RGB-D sequences for indoor tracking, evaluated at stride 3 and stride 10. **NRGBD** contains diverse indoor scenes, also evaluated at both stride settings.

For point cloud reconstruction, we report Chamfer Distance (CD), Accuracy (Acc), Completeness (Comp), and Normal Consistency (NC). For camera pose estimation, we measure Absolute Trajectory Error (ATE), Absolute Rotation Error (ARE), and Relative Pose Errors for rotation (RPE-rot) and translation (RPE-trans).

FastVGGT is implemented on top of VGGT [19] and requires no retraining or fine-tuning. All experiments are conducted on a single NVIDIA A100 GPU. We compare against several baselines: VGGT* (our memory-efficient variant of VGGT that discards unused intermediates), Fast3R [23], CUT3R [20], π^3 [22], and StreamVGGT [25]. Unless otherwise noted, FastVGGT uses a merging ratio of 0.9 with token merging applied across all transformer blocks.

4.2. Point Cloud Reconstruction

ScanNet-50. Table 1 presents the reconstruction results on ScanNet-50. Both π^3 and StreamVGGT fail with out-of-memory errors across all sequence lengths, demonstrating that existing methods cannot scale to long input sequences on a single GPU. Fast3R and CUT3R succeed but produce substantially worse reconstructions, with CD values of 0.684 and 0.786, respectively, at 1000 images. In contrast, FastVGGT achieves a CD of 0.425, outperforming even the VGGT* baseline (0.471) while running approximately $4\times$ faster (180.7s vs. 724.6s). This improvement persists across all sequence lengths: at 500 images, FastVGGT achieves CD 0.411 vs. 0.420 with a $3.2\times$ speedup, and at 300 images it matches the baseline CD of 0.416 while being $5.5\times$ faster. The accuracy improvement at longer sequences suggests that token merging acts as an implicit regularizer, reducing the influence of redundant and potentially misleading tokens in global attention.

7 Scenes. Table 2 shows results on the 7 Scenes dataset. At stride 3, FastVGGT achieves an accuracy of 0.018 (vs. VGGT* 0.019), completeness of 0.026 (vs. 0.027), and normal consistency of 0.617 (vs. 0.611), while reducing inference time from 76.7s to 28.0s—a $2.7\times$ speedup. At stride 10, similar trends hold with FastVGGT running in 5.1s compared to 8.7s for the baseline. Again, π^3 and StreamVGGT encounter OOM errors, while CUT3R shows poor accuracy at stride 3 (Acc = 0.179).

NRGBD. As shown in Table 3, FastVGGT maintains comparable accuracy to VGGT* on the NRGBD dataset (Acc = 0.029 for both at stride 3), while achieving higher normal consistency (0.730 vs. 0.727) and a $2.6\times$ speedup (53.1s vs. 136.1s). At stride 10, FastVGGT reduces inference time from 13.9s to 7.3s while preserving reconstruction quality.

4.3. Camera Pose Estimation

Table 4 presents pose estimation results on ScanNet-50. FastVGGT consistently improves over the VGGT* baseline, with the gains most pronounced at longer sequences. At 1000 frames, FastVGGT reduces ATE from 0.196 to 0.164 (a 16.3% improvement), ARE from 4.636 to 3.860, RPE-rot from 0.997 to 0.667 (a 33.1% reduction), and RPE-trans from 0.039 to 0.029. At 500 frames, ATE improves from 0.174 to 0.145. The improvements diminish at shorter sequences (300 and 100 frames), where error accumulation is less severe.

These results indicate that token merging mitigates the error drift that arises in long-sequence pose estimation. As the number of frames grows, dense global attention increasingly attends to redundant and noisy tokens, amplifying small per-frame errors into large trajectory drift. By merging redundant tokens, FastVGGT reduces this noise propagation while preserving geometrically informative features.

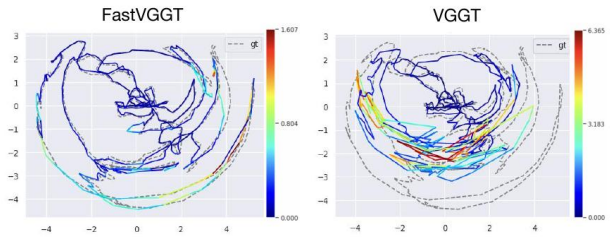

Figure 5. Qualitative comparison of camera pose estimation on ScanNet-50 (1000 frames). FastVGGT (blue) maintains better alignment with the ground truth compared to VGGT* (red), demonstrating reduced long-sequence drift.

Fig. 5 visualizes this effect: the VGGT* trajectory diverges progressively from ground truth over 1000 frames, while FastVGGT maintains closer alignment throughout.

4.4. Efficiency Analysis

Beyond reconstruction quality, we examine the memory and throughput characteristics of FastVGGT across sequence lengths. VRAM consumption in VGGT* scales quadratically with the number of input frames; at 1,000 frames it approaches the memory ceiling of a single A100 GPU. FastVGGT reduces peak VRAM usage by roughly $3\times$ through aggressive token merging, enabling inference on hardware configurations where VGGT* would otherwise fail. The speedup profile is not uniform across sequence lengths: at 100 frames, global attention is not yet the dominant cost, so the benefit is modest. As the sequence length grows, the quadratic attention term increasingly dominates and the fixed overhead of token partitioning and unmerging becomes negligible by comparison. This explains why the speedup peaks at 300 frames ($5.5\times$) before stabilizing near $4\times$ at 1,000 frames. Taken together, these scaling properties confirm that FastVGGT's efficiency gains are most pronounced in the long-sequence regime where scalability is the critical practical constraint.

4.5. Ablation Studies

Token Partitioning Strategies. Table 5 evaluates the contribution of each component in our 3D-aware token partitioning scheme on ScanNet-50 (500 images). Row (a) applies token merging with purely random src/dst assignment, yielding poor CD (0.947) and ATE (0.842)—merging without structure destroys geometric correspondences. Adding uniform spatial sampling (b) improves CD to 0.637 but ATE remains high (0.722), as the reference frame anchor is missing. Introducing reference frame preservation (c) dramatically improves ATE to 0.149, confirming that the first frame serves as a critical geometric anchor for consistent world-frame alignment. Finally, adding salient token preservation (d) yields the best overall results (CD = 0.411, ATE

Method	1000		500		300		100	
	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓
π^3 [22]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT[25]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [23]	0.684	397.8s	0.701	97.3s	0.711	34.9s	0.723	4.8s
CUT3R [20]	0.786	34.8s	0.774	18.8s	0.775	11.1s	0.767	3.6s
VGGT* [19]	0.471	724.6s	0.420	177.5s	0.416	131.4s	0.423	9.1s
FastVGGT	**0.425**	**180.7s**	**0.411**	**55.2s**	**0.416**	**23.8s**	**0.426**	**5.4s**

Table 1. Quantitative results of point cloud reconstruction on the ScanNet-50 dataset with input sequences of 1000, 500, 300, and 100 images. *OOM* denotes out-of-memory.

Method	7 Scenes - Stride 3							7 Scenes - Stride 10						
	Acc ↓		Comp ↓		NC ↑		Time ↓	Acc ↓		Comp ↓		NC ↑		Time ↓
	Mean	Med.	Mean	Med.	Mean	Med.		Mean	Med.	Mean	Med.	Mean	Med.	
π^3 [22]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT [25]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [23]	0.045	0.027	0.047	0.010	0.616	0.627	43.7s	0.040	0.021	0.056	0.013	0.639	0.657	5.5s
CUT3R [20]	0.179	0.121	0.097	0.043	0.588	0.581	14.5s	0.041	0.021	0.029	0.010	0.651	0.677	4.2s
VGGT* [19]	0.019	0.008	0.027	0.010	0.611	0.628	76.7s	0.020	0.008	0.027	0.010	0.623	0.641	8.7s
FastVGGT	**0.018**	**0.008**	**0.026**	**0.010**	**0.617**	**0.634**	**28.0s**	**0.018**	**0.008**	**0.027**	**0.010**	**0.628**	**0.648**	**5.1s**

Table 2. Quantitative results of point cloud reconstruction on the 7 Scenes dataset. Stride denotes keyframes sampled every 3 or 10 frames.

= 0.141), demonstrating that retaining high-importance tokens further preserves fine-grained geometric details.

Merging Ratio and Affected Blocks. Table 6 analyzes the effect of the merging ratio and the number of transformer blocks where merging is applied. When merging is applied across all blocks (Blocks = 0), a ratio of 0.9 achieves the best trade-off between quality (CD = 0.411) and speed (55.2s). Reducing the ratio to 0.3 or 0.6 yields marginal quality improvements (CD = 0.408 and 0.415) but at 2× or more the inference time. Restricting merging to a subset of blocks (10 or 20) increases both computation time and reconstruction error, as partial merging creates inconsistencies across layers. These results justify our default configuration of merging in all blocks with ratio 0.9.

Naive Sampling Baselines. Table 7 compares our 3D-aware partitioning against naive random and fixed-stride token selection strategies. Random sampling with ratio 0.5 achieves CD = 0.32 but is slower (22.6s) due to lower merging ratios. At ratio 0.8, quality degrades to CD = 0.44. Fixed-stride selection with stride 5 yields CD = 0.33, while stride 8 produces severe degradation (CD = 0.77). All naive strategies substantially underperform the unmerged baseline (CD = 0.21), whereas our full method (Table 1) achieves strong quality at much higher merging ratios. This demonstrates that 3D-aware token partitioning—preserving reference frames and salient tokens—is essential for maintaining reconstruction fidelity under aggressive token merging.

5. Conclusion

We present FastVGGT, a training-free acceleration method for the Visual Geometry Transformer that enables scalable 3D reconstruction and camera pose estimation from long image sequences. By exploiting inherent token redundancy in VGGT's global attention via 3D-aware token merging, FastVGGT achieves up to 4× speedup while maintaining or improving reconstruction accuracy and pose estimation quality. Our key design choices—reference frame anchoring, salient token preservation, and region-based uniform sampling—ensure that geometric consistency is preserved despite aggressive token reduction. Notably, FastVGGT not only accelerates inference but also mitigates error drift in long-sequence pose estimation by reducing the influence of redundant tokens in global attention. Experiments across ScanNet-50, 7 Scenes, and NRGBD validate these gains, demonstrating that principled training-free token merging is

Method	NRGBD - Stride 3							NRGBD - Stride 10						
	Acc ↓		Comp ↓		NC ↑		Time ↓	Acc ↓		Comp ↓		NC ↑		Time ↓
	Mean	Med.	Mean	Med.	Mean	Med.		Mean	Med.	Mean	Med.	Mean	Med.	
π^3 [22]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT [25]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [23]	0.074	0.036	0.024	0.011	0.658	0.682	68.9s	0.061	0.028	0.031	0.013	0.669	0.712	7.4s
CUT3R [20]	0.346	0.243	0.184	0.090	0.579	0.623	18.3s	0.132	0.064	0.056	0.0106	0.669	0.725	5.7s
VGGT* [19]	0.029	0.019	0.018	0.009	0.727	0.728	136.1s	0.016	0.010	0.017	0.009	0.735	0.738	13.9s
FastVGGT	**0.029**	**0.021**	**0.019**	**0.010**	**0.730**	**0.738**	**53.1s**	**0.018**	**0.011**	**0.018**	**0.009**	**0.736**	**0.741**	**7.3s**

Table 3. Quantitative results of point cloud reconstruction on the NRGBD dataset.

Input	ATE ↓		ARE ↓		RPE-rot ↓		RPE-trans ↓	
Frames	Baseline	Ours	Baseline	Ours	Baseline	Ours	Baseline	Ours
1000	0.196	**0.164**	4.636	**3.860**	0.997	**0.667**	0.039	**0.029**
500	0.174	**0.145**	4.190	**3.591**	0.963	**0.627**	0.042	**0.031**
300	0.145	**0.142**	3.689	**3.554**	0.786	**0.801**	0.040	**0.036**
100	0.140	**0.141**	3.625	**3.512**	1.224	**1.262**	0.058	**0.061**

Table 4. Quantitative results of camera pose estimation on the ScanNet-50 dataset.

Methods	Uniform	Reference	Salient	CD ↓	ATE ↓
(a)	-	-	-	0.947	0.842
(b)	✓	-	-	0.637	0.722
(c)	✓	✓	-	0.431	0.149
(d)	✓	✓	✓	0.411	0.141

Table 5. Ablation study for Token partitioning strategies.

Blocks	0.3		0.6		0.9	
	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓
0	0.408	119.8s	0.415	64.3s	0.411	55.2s
10	0.418	146.2s	0.424	118.4s	0.431	106.3s
20	0.423	172.9s	0.427	169.1s	0.411	157.3s

Table 6. Ablation study on merging ratio and affected blocks.

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	Random		Fixed		Baseline
	$r = 0.5$	$r = 0.8$	$s = 5$	$s = 8$	
CD ↓	0.32	0.44	0.33	0.77	0.21
Time ↓	22.6	16.9	17.2	11.6	30.4

Table 7. Results of applying merging strategies based on random sampling and fixed-stride sampling. Here, r denotes the merging ratio, and s represents the stride for selecting dst tokens.

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